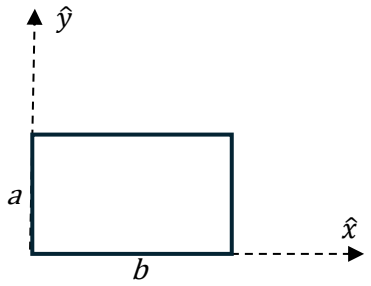
Physics 2 Problem Solving Video – Part C Gauss' Law

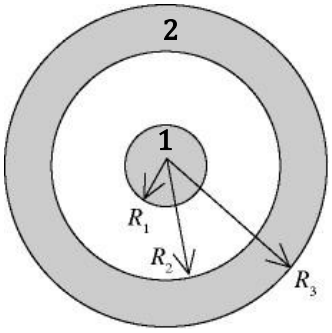
1) A rectangular shape is in the x-y plane as shown in the figure below, If the electric field strength has a y-axis dependency and points along the z-axis, determine the electric flux through the rectangle. $\hat{z}$ points out of the page and assume m is a constant.

$$\vec{E}(y) = (my^3)\hat{z}$$



Answer: $\Phi = \frac{mba^4}{4}$

2) Two concentric spheres are shown in the figure. The inner sphere (1) is a solid conductor and carries a charge of 10.0 μC positive, uniformly distributed over its outer surface. The outer sphere (2) is a conducting shell, a negative charge of 2.0 μC is placed on the outer shell. No other charges are present. The radii shown in the figure have the values R_1 = 2.0 cm, R_2 = 10.0 cm, and R_3 = 12.0 cm.



A) Find the total excess charge on the inner and outer surfaces of the conducting sphere.

$q_{inner} = 10\ \mu C\ negative \quad q_{outer} = 8\ \mu C\ Postive$

B) Find the magnitude and direction of the electric field at the following distances *r* from the center of the inner sphere:

(i) *r* = 1.0 cm, *Anwer: 0 N/C*

(ii) *r* = 5.0 cm, *Anwer: 3.6 x 10^7 N/C*

(iii) *r* = 11.0 cm, *Anwer: 0 N/C*

(iv) *r* = 15.0 cm. *Anwer: 3.2 x 10^6 N/C*

3) A spherical, non-conducting object with radius R = 20 cm carries a total charge Q = 10 nC distributed uniformly throughout the volume of the sphere.

   A) Determine the magnitude of the electric field at a distance r = 10 cm from the center of the sphere?

      Anwer: 1124 N/C

   B) Determine the magnitude of the electric field at a distance r = 40 cm from the center of the sphere?

      Anwer: 2248 N/C

4) A conducting cylinder of length L = 1 m has a radius of 10 cm and a surface charge density of $\sigma = 5 \times 10^{-8}$ C/m^2, determine the electric field strength 40 cm above the cylinder using Gauss's Law.

   Anwer: 1412 N/C